Ioannis Balasis

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Research Interests

Deep learning for medical image analysis (MRI, digital pathology), explainable and trustworthy AI, computational oncology, and multimodal integration of imaging and clinical data for precision medicine.

Education

Diploma in Computer Engineering and Informatics 2020 – 2026
University of Patras *Patras, Greece*
Integrated Master's degree (5-year programme, 300 ECTS) · Grade: 6.72/10
Thesis: *Machine and Deep Learning Techniques for Brain Tumor Detection and Classification Using MRI*

Publications

I. Balasis, C. Makris. "Deep Learning Architectures for Brain Tumor MRI Classification: Systematic Comparison with a Novel Hybrid CNN-Transformer, Explainable AI and Statistical Significance Testing." *Applied Intelligence* (Springer Nature). **Under review**, 2026.

Research Experience

Erasmus+ Research Trainee – Computational Pathology Sep 2026 – Feb 2027 (exp.)
Karolinska Institutet, Department of Oncology-Pathology *Stockholm, Sweden*
Supervisors: Prof. Johan Hartman, Dr. Balázs Ács

- AI-based digital pathology research in breast cancer, focusing on image-based biomarkers and analysis of the tumor immune microenvironment.
- Whole-slide image analysis: stain normalization, quality control, and quantification of tumor-infiltrating lymphocytes (TILs) and spatial features.
- Data curation and automated processing of large clinical whole-slide image cohorts.
- Deep learning methods for computational pathology, including foundation models and multiple-instance learning (MIL).
- Close collaboration with pathologists and researchers in an interdisciplinary clinical research environment.

Master's Thesis Research Mar 2025 – Jun 2026
University of Patras *Patras, Greece*

- Systematic comparison of nine deep learning architectures (custom CNNs, transfer learning, Vision Transformers) for four-class brain tumor MRI classification.
- Novel TensorFlow implementation of MedViT V2 (KAN-integrated hybrid CNN–Transformer) and a feature-fusion dual-branch extension.
- Rigorous non-parametric statistical framework across all models (Friedman, Nemenyi, Wilcoxon with Bonferroni correction, Cohen's d).
- Multi-method explainability analysis (Grad-CAM, SHAP, Integrated Gradients, Occlusion) revealing an architecture-agnostic failure pattern on meningioma (the "Meningioma Paradox").
- Clinical deployment prototype (Streamlit) with consensus voting across architectures and automated PDF reporting.
- Best custom CNN: 98.60% accuracy; MedViT V2 Dual: 98.15% (statistically equivalent, $p = 0.240$). Resulted in a manuscript under review.

Dry Lab Member Dec 2024 – Dec 2025
Patras Medicine iGEM 2025 Team, University of Patras *Patras, Greece*

- Computational models of genetic circuits and metabolic pathways with ODE-based and stochastic simulations (COPASI, iBioSim, SimBiology); stability and sensitivity analyses.

- Molecular docking and protein modeling (SwissDock, AlphaFold, UCSF Chimera) to assess candidate therapeutic interactions.
- Worked closely with the wet lab to align computational predictions with experimental design.
- Presented the project *Morphe* at the iGEM Grand Jamboree, Paris.

Professional Experience

Healthcare IT Intern Mar 2026 – May 2026
Siemens Healthineers *Athens, Greece*

- Developed a digital pathology tool automating the conversion of whole slide images (NDPI) to DICOM and their transmission to Syngo Carbon IDM via the DICOM C-STORE protocol; **deployed in a live hospital environment**.
- Built it as a standalone Windows application (.exe) using Python, pydicom, pynetdicom and OpenSlide.
- Analysed and extracted clinical data from hospital databases using Oracle SQL and Microsoft SQL Server.
- Authored full technical documentation (LaTeX, Word) and published the source code on GitHub.

Dry Lab Advisor (Mentor) Jan 2026 – Present
Patras Medicine iGEM 2026 Team *Patras, Greece*

- Mentoring undergraduate researchers on computational modeling, machine learning and in silico experimental design.

Laboratory Teaching Assistant, Basic Electronics Sep 2022 – Feb 2023
University of Patras *Patras, Greece*

Honors and Awards

- **iGEM 2025** (Paris): Gold Medal; Best Therapeutics Project (Undergraduate); Top 10 Undergraduate Team.
  Nominations: Best Wiki, Best Project Presentation, Best New Composite Part, Best Sustainable Development Impact.
- **Harvard CS50x Puzzle Day**: 1st Place (2023).

Technical Skills

Medical imaging	MRI, NIfTI, preprocessing, segmentation (U-Net), classification, explainable AI
Digital pathology	Whole-slide images, stain normalization, TIL quantification, MIL, foundation models, OpenSlide, DICOM (pydicom, pynetdicom)
Deep learning	PyTorch, TensorFlow, Keras, CNNs, Vision Transformers, transfer learning
Scientific computing	NumPy, SciPy, scikit-learn, statistical testing, ODE/stochastic modeling
Programming	Python, C, Java, R, SQL (Oracle, MS SQL Server), HTML/CSS/JavaScript

Training and Certifications

- BCI & Neurotechnology Spring School, g.tec medical engineering GmbH (2025)
- Masterclass: Python for Neuroscience, g.tec medical engineering GmbH (2025)

Languages

Greek (native) · English (advanced) · German (intermediate)

References

Available upon request.